

flow (2), this is by the relation

$$\log p_t(x(t)) - \log p_s(x(s)) = - \int_s^t \nabla \cdot v(x(\tau), \tau) d\tau,$$

which involves the time-integration of the trace of the Jacobian of v_t [29]. While these computations may encounter challenges in high dimensions, the ability to evaluate the (log) likelihood is fundamentally useful; in particular, it allows for evaluating the maximum likelihood training objective on finite samples. This property is also adopted in the deterministic reverse process in diffusion models [67], called the “probability flow ODE” (see more in Section 1.2.1), so the likelihood can be evaluated once a forward diffusion model has been trained.

Progressive flow models. Another line of works, developed around the same time as diffusion models, explored the variational form of normalizing flow as a Wasserstein gradient flow and proposed the so-called *progressive* training of the flow model. The progressive training of ResNet, i.e., training block-wise by a per-block variational loss, was proposed by [38] at an earlier time under the GAN framework. Later on, the Jordan-Kinderlehrer-Otto (JKO) scheme, as a time-discretized Wasserstein gradient flow (see more in Section 2.3), was explored in several flow-based generative models: [4, 54] implemented the JKO scheme using input convex neural networks [8]; [26] proposed a forward progressive flow from noise to data, showing empirical success in generating high-dimensional real datasets; [74] developed the JKO flow model under the invertible continuous-time CNF framework, achieving competitive generating performance on high-dimensional real datasets at a significantly reduced computational and memory cost from previous CNF models; an independent concurrent work [70] proposed a block-wise JKO flow model utilizing the framework of [58]. Many other flow models related to diffusion and Optimal Transport (OT) exist in the literature; see more in Section 1.2.2. Our theoretical analysis will focus on the progressive flow models, and we primarily follow the invertible flow framework in [74].

To be more specific, a progressive flow model represents the flow on $[0, T]$ as the composition of N sub-flow models where each one computes the flow on a sub-interval $[t_{n-1}, t_n]$, $n = 1, \dots, N$. The training is “progressive”, meaning that at one time, only one sub-model is trained, and the training of the n -th sub-model is conducted once the previous $n - 1$ sub-models are trained and fixed. The progressive block-wise training is in contrast of the end-to-end training, where the flow on $[0, T]$ (or N Residual Blocks) is trained simultaneously by a single objective. The sub-flow model on $[t_{n-1}, t_n]$ can take different forms, e.g., a ResNet block or a continuous-time neural ODE, and the N sub-intervals always provide a time-discretization of the flow. In this context, we call the sub-model on the n -th sub-interval a “Residual Block”.

1.2 Additional related works

1.2.1 Score-based diffusion models

In score-based diffusion models, the algorithm first simulates a forward process, which is a (time-discretized) SDE, from which a score function parameterized as a neural network is trained. The reverse (SDE or ODE) process is simulated using the learned score model to generate data samples.

SDE in diffusion models As a typical example, in the *variance preserving* Denoising Diffusion Probabilistic Modeling (DDPM) process [33, 67], the forward process produces a sequence of X_n ,

$$X_n = \sqrt{1 - \beta_n} X_{n-1} + \sqrt{\beta_n} Z_{n-1}, \quad n = 1, \dots, N, \quad (4)$$

where $Z_n \sim \mathcal{N}(0, I_d)$ i.i.d. and $X_0 \sim P$ is drawn from data distribution. With large N , the continuum limit of the discrete dynamic (4) is a continuous-time SDE,

$$dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)}dW_t, \quad t \in [0, T], \quad (5)$$

where $\beta(t) > 0$ is a function and W_t is a standard Wiener process (Brownian motion). Since $\beta(t)$ in (5) corresponds to a time reparametrization of t , after changing the time ($t \mapsto \int_0^t \beta(s)/2ds$), (5) becomes the following SDE

$$dX_t = -X_t dt + \sqrt{2}dW_t, \quad (6)$$

which is the Ornstein-Uhlenbeck (OU) process in $\mathbb{R}^d$. We consider the time-parametrization in (6) for exhibition simplicity. More generally, one can consider a diffusion process

$$dX_t = -\nabla V(X_t) dt + \sqrt{2}dW_t, \quad X_0 \sim P, \quad (7)$$

and the OU process is a special case with $V(x) = \|x\|^2/2$.

We denote by $\rho_t = \mathcal{L}_t(P)$ the marginal distribution of X_t for $t > 0$. The time evolution of ρ_t is described by the Fokker-Planck Equation (FPE) written as

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla V + \nabla \rho_t). \quad (8)$$

Forward and reverse processes When simulating the forward process, the diffusion models train a neural network to learn the score function $s_t(x) := \nabla \log \rho_t$ by score matching [36, 71]. The training objective can be expressed as the mean-squared error defined as $\int_0^T \int \|\hat{s}_t(x) - s_t(x)\|^2 \rho_t(x) dx dt$, which facilitates training and is scalable to high dimension data such as images (in the original pixel space).

Once the neural-network score function $\hat{s}_t$ is learned, the algorithm simulates a reverse-time SDE $\tilde{X}_t$ (with time discretization in practice) [67], such that from $\tilde{X}_T \sim \mathcal{N}(0, I)$ the distribution of $\tilde{X}_0$ is expected to be close to the data distribution P . It has also been proposed in [67] to compute the reverse process by integrating the following ODE reverse in time

$$\dot{\tilde{x}}(t) = -\nabla V(\tilde{x}(t)) - s_t(\tilde{x}(t)), \quad (9)$$

and (9) was called the “probability flow ODE.” The validity of this ODE reverse process can be justified by the observation that the CE (3) and FPE (8) are the same when setting $v_t(x) = -(\nabla V(x) + s_t(x))$. This equivalence between density evolutions by SDE and ODE has been known in the literature of diffusion processes and solving FPE, dating back to the 90s [21, 22].

1.2.2 Flow models related to diffusion and OT

Flow-matching models After diffusion models gained popularity, several flow-based models (in the reverse and forward directions) closely related to the diffusion model emerged. In particular, the Flow-Matching ODE model was proposed in [50] using the formulation conditional probability paths, where a neural ODE parameterized $\hat{v}(x, t)$ is trained to match a velocity field $v(x, t)$ whose corresponding CE (3) can evolve the density p_t towards normality. The algorithm can adopt diffusion paths, where the CE will equal the density evolution equation (8) of an SDE forward process, as well as non-diffusion paths. A similar approach was developed under the “stochastic interpolant” framework in [3], where the terminal distribution q can be arbitrary (not necessarily the normal distribution) and only accessible via finite samples. These models train a continuous-time CNF by minimizing a “matching” objective instead of the maximum likelihood objective as in [29], thus avoiding the computational challenges of the latter.

Optimal Transport flows Apart from diffusion models and Wasserstein gradient flow, Wasserstein distance and OT have inspired another line of works on flow models where the Wasserstein distance, or a certain form of transport cost, is used to regularize the flow model and to compute the OT map between two distributions. Transport cost regularization of neural network models was suggested in several places: [62] provided a general framework for solving high-dimensional mean-field games (MFG) and control problems, [27] proposed a kinetic regularization aiming to stabilize neural ODE training, [58] and [72] developed the transport regularization in CNF and invertible ResNet, respectively, and [34] applied to MFG and flow models. Other works developed flow models to compute the optimal coupling or the optimal transport between two distributions. For example, Rectified Flow [51] proposed an iterative method to adjust the flow towards the optimal coupling. The method is closely related to the stochastic interpolant approach [3] which, in principle, can solve the OT trajectory if the interpolant map can be optimized. A flow model to compute the dynamic OT between two high dimensional distributions from data samples was proposed in [73] by refining the flow using the transport cost from a proper initialization. Despite the wealth of methodology developments and empirical results, the theoretical guarantees of these flow models are yet to be developed.

1.2.3 Theoretical guarantees of generative models

Approximation and estimation of GAN On theoretical guarantees of generative models, earlier works focused on the approximation and estimation analysis under the GAN framework. The expressiveness of a deep neural network to approximate high dimensional distributions was established in a series of works, e.g., [46, 52, 60, 75], among others. The neural network architectures in these universal approximation results are typically feed-forward, like the generator network (G-net) proposed in the original GAN. The approximation and estimation of the discriminator network (D-net) in GAN were studied in [19], and the problem can be cast and analyzed as the learning of distribution divergences in high dimension [68]. Convergence analysis of GAN was studied in several places, e.g., [35].

Guarantees of diffusion models An earlier work [69] studied the expressiveness of a generative model using a latent diffusion process and proved guarantees for sampling and inference; however, the approach only involves a forward process and differs from the recent diffusion models. Motivated by the prevailing empirical success of score-based diffusion models, recent theoretical works centralized on the generation guarantee of such models using both SDE and ODE samplers, i.e., the reverse process.

For the SDE reverse process, the likelihood guarantee of the score-based diffusion model was first derived in [65] without time discretization. Taking into account the time discretization, which significantly influences the efficiency in practice, a series of theoretical studies have established polynomial convergence bounds for such models [11, 14, 17, 20, 47, 48, 59]. In particular, [47] were the first to attain polynomial convergence without succumbing to the curse of dimensionality, although this required a log-Sobolev Inequality on the data distribution. For a general data distribution, [17] achieved polynomial error bounds in Total Variation (TV) distance under the Lipschitz assumption, leveraging Girsanov’s theorem. In parallel, [48] derived similar polynomial convergence bounds, employing a technique for converting L^∞ -accurate score estimates into L^2 -accurate score estimation. Further advancements by [14] established a more refined bound, reducing the requirement of smoothness of data distribution. Most recently, [59] improved the convergence rates under mild assumptions by introducing prediction-correction, and [11] established the first convergence bounds for diffusion models, which are linear in the data dimension (up to logarithmic factors) without